

Jin Kweon, MSc

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SUMMARY

Bioinformatician with cross-domain expertise spanning clinical (i.e. REDCap) and -omics data across oncology, immunology, and neurodegeneration in both industry and academic settings. Builds scalable pipelines, web-based tools, and published statistical deep learning models (ISMB, IEEE, ASHG) in R and Python.

EDUCATION

McGill University

MSc, Experimental Medicine – GPA: 4.00

Thesis: “DTractor: Advancing cell type deconvolution of spatial transcriptomics with deep neural network, transfer learning and matrix factorization”

Montreal, QC

Jan 2023 – Dec 2024

McGill University (Military service: 2019 – 2021)

PhD Candidate, Quantitative Life Science Withdrawn

Montreal, QC

Sep 2018 – Dec 2021

University of California, Berkeley

Bachelor of Arts, Statistics (Focus: Mathematics / Economics) – Major GPA: 4.00

Berkeley, CA

Aug 2016 – May 2018

- **Relevant Coursework:** Applied Machine Learning, Structure and Interpretation of Computer Programs, Multivariate Statistics, Linear Algebra, Linear Modeling, Statistical Bioinformatics, Genomics for drug discovery and development

WORK EXPERIENCE

Copoly.ai (Genentech)

Computational Biologist

Toronto, ON

May 2025 – Present

- Designed a differential-correlation approach to flag disease-linked co-abundance shifts between cell neighborhoods at single-cell resolution, addressing a blind spot in CNA and Milo, which evaluate neighborhoods in isolation (ASHG).
- Tested interactions across cellular neighborhoods with genetics and spatial integration, revealing coordinated disease responses in AMD, IBD, AD, IPF, SLE, including enterocyte-immune signatures in Crohn's.
- Built a custom significance-testing pipeline benchmarked against parametric approximation methods, controlling error rates across millions of neighborhood-pair comparisons.
- Solved for compositional bias across samples by introducing compartment-aware, log-scaled normalization ahead of comparison.
- QC iPSC-derived RPE differentiation batches by combining single-cell-level annotation with expression and pathway analysis to flag off-target cell lines before cell-therapy release.

Montreal General Hospital

Data Analyst

Montreal, QC

Feb 2025 – May 2026

- Tackled missing-data gaps in oncology PRO data by benchmarking a dozen approaches — deep learning (VAE, DAE, LSTM), transformers (Gemma, TimesFM), ensembling — paired with HuggingFace-based FACT scoring for imputation (IEEE EMBC).
- Merged three clinical data sources to establish preoperative emotional distress as a driver of longer hospital stays and readmission risk after esophagectomy, recognized with an ESDE 2025 award.
- Revealed in multivariable models that preoperative symptom burden predicted quality-of-life improvement one year post-esophagectomy, with smoking showing a possible trend, while tumor characteristics, surgical approach, and neoadjuvant treatment showed no association; female sex independently predicted reduced improvement.
- Tracked textbook-outcome achievement across a 228-patient, multi-year cohort as a marker of both surgical quality and long-term (disease-free/overall) survival.

Douglas Research Center

Bioinformatician

Montreal, QC

Mar 2025 – May 2025

- Ported and extended meffil/ewaff (originally R) into Python, adding non-linear CpG-phenotype modeling for differentially methylated positions detection (roughly doubling predictive accuracy).
- Built a spline-based method that outperformed Enmix combp and bumhunter at catching sex-specific methylation regions/sites across tissue types and age groups that those tools missed.
- Layered interaction and ensemble modeling on top to separate sex-dependent from sex-independent methylation effects, and to distinguish tissue-specific vs. tissue-consistent age-related changes.

Research on developing cell type deconvolution algorithms

Montreal, QC

Computational Biologist (spatial transcriptomics project)

Jan 2023 – Dec 2024

- Implemented the *DTractor* software that demonstrated higher accuracy, resolution, robustness, efficacy, and scalability on two simulated datasets and three real datasets, outperforming eight other popular and commonly used deconvolution methods (accepted ISMB MLCBS).
- Developed a method that uses deep neural network training on sc/snRNA-seq and spatial transcriptomics (ST) data, incorporating transfer learning from scRNA-seq reference data to ST data, and conducting iterative matrix factorizations.
- Decomposed ST at cell-type resolutions to gain insights into spatial organization in pancreatic adenocarcinoma, uncovering tumor microenvironment interactions and offering rare capability to estimate cell counts per location.

Research on developing AI drug design-related algorithms

Montreal, QC / Korea

Research Analyst (single cell project)

Sep 2021 – Nov 2023

- Introduced *scBeacon*, an innovative framework built upon a VQ-VAE framework, deep contrastive siamese network, and a greedy iterative strategy, to effectively pinpoint differential genes and identify the same cell population across different biological conditions forming cluster pairs.
- Advanced biomarker discovery by liberating differential gene analysis from manual annotation constraints, enabling *scBeacon* to identify critical cluster pairs and potential therapeutic targets in COVID-19 datasets.
- Demonstrated therapeutic response by mapping cytarabine-induced cellular adaptations using optimal embedding vector quantization and probabilistic models, revealing elusive drug resistance pathways in treatment populations.

Adecco (Apple)

Singapore / Korea

Apple Map Data Analyst

Jan 2022 – Dec 2022

- Evaluated map navigation, search, and suggestions on iOS/macOS through hundreds of statistical tests (linear, non-linear, logistic regression), tying user query patterns and geography to output accuracy.
- Diagnosed root causes behind 200+ flagged map-data problems using hypothesis testing and causal analysis (a 20% drop in navigation errors followed).
- Set up scraping/extraction-driven benchmarks to systematically score map reliability across address precision, POI correctness, metadata accuracy, and relevance.

SKILLS

- Proficient in R [seurat], Python [pytorch, tensorflow, scanpy]. Basic knowledge of Java, SQL, CSS, HTML, MATLAB, Linux, C++, HPC/cloud computing, Git

PUBLICATIONS & ACKNOWLEDGEMENTS

- **Yong Jin Kweon** et al. (2025). Comparing Missing Data Imputation Methods for Patient-Reported Outcomes in Esophageal Cancer Research. <https://www.medrxiv.org/content/10.1101/2025.09.10.25335531v2>
- Samir S. Amin, **Jin Kweon**, Joshua S. Cheruvathur, Mehrnosh Dehghani, Sara Najmeh, Carmen Mueller, Jonathan Cools-Lartigue, Lorenzo Ferri, R. Trafford Crump (2025). Preoperative Quality-of-Life and Recovery Following Esophagectomy: Identifying Modifiable Factors Associated with Improvement. <https://www.medrxiv.org/content/10.1101/2025.07.30.25332378v1>
- **Yong Jin Kweon**, Chenyu Liu, Gregory Fonseca, Jun Ding (2025). DTractor enhances cell type deconvolution in spatial transcriptomics by integrating deep neural networks, transfer learning, and matrix factorization. <https://www.biorxiv.org/content/10.1101/2025.04.12.648541v1>
GitHub: <https://github.com/mcgilldinglab/DTractor>
- Chenyu Liu, **Yong Jin Kweon**, Jun Ding (2023). scBeacon: single-cell biomarker extraction via identifying paired cell clusters across biological conditions with contrastive siamese networks. <https://arxiv.org/abs/2311.02594>
GitHub: <https://github.com/mcgilldinglab/scBeacon>
- Yuying Lu, Camille Morencé, **Yong Jin Kweon**, Archer Yi Yang, Jun Ding (2025). SC2CD Co-optimizes Spatial Clustering and Cell Decomposition through Bidirectional Information Flow in Single-Cell Spatial Transcriptomics.
- Haber, J.R. (2020). Sorting Schools: A Computational Analysis of Charter School Identities and Stratification. Sociology of Education. <https://doi.org/10.1177/0038040720953218>

PRESENTATIONS

- Yousif Salman, Emad Mohammed, **Yong Jin Kweon**, Shayan Dhillon, Trafford Crump. (2026). Context-Aware Prompt Engineering for LLM-Based Missing Data Imputation in Patient-Reported Outcomes. IEEE Engineering in Medicine and Biology Society. Toronto, Canada. July 26-30, 2026.
- Xiuhui Yang, Jingtao Wang, **Yong Jin Kweon**, Chenyu Liu, Gregory Joseph Fonseca, Jun Ding. (2026). DTractor: Dual-reference modeling enables biologically coherent deconvolution of spatial transcriptomics. ISMB MLCSB. Washington, D.C., USA. July 12-16, 2026.
- **Jin Kweon**, Trafford Crump. (2025). Comparing Missing Data Imputation Methods for Patient-Reported Outcomes in Esophageal Cancer Research. Digital Health Symposium: From Research Infrastructure to Clinical Care. Montreal, Canada. Nov 4, 2025.
- Benjamin Tordjman, **Jin Kweon**, Nataliya Burban El Dirani, Mehrnoush Dehghani, Mathieu Rousseau, Carmen Mueller, Sara Najmeh, Jonathan Spicer, Jonathan Cools-Lartigue, Lorenzo Ferri, Trafford Crump. (2025). Evaluation of textbook outcomes and long-term prognosis following esophagectomy: a retrospective cohort study. International Society for Disease of the Esophagus (ISDE) World Congress. Brisbane, Australia. Sep. 18-20, 2025.
- Nisha Suarez, **Jin Kweon**, Mehrnoush Dehghani, Lorenzo Ferri, Sara Najmeh, Carmen Mueller, Jonathan Cools-Lartigue, Jonathan Spicer, Trafford Crump. (2025). Testing the effect of changes in emotional distress on post-esophagectomy outcomes. European Society for Diseases of the Esophagus (ESDE) conference. Amsterdam, Netherlands, May. 12, 2025. DOI: 10.17605/[OSF.IO/UWBN2](https://osf.io/UWBN2)
- Samir Amin, Joshua Cheruvathur, **Jin Kweon**, Mehrnoush Dehghani, Sara Najmeh, Carmen Mueller, Jonathan Cools-Lartigue, Lorenzo Ferri, R. Trafford Crump. (2025). Rethinking Recovery Predictors in Esophageal Cancer Surgery. European Society for Diseases of the Esophagus (ESDE) conference. Amsterdam, Netherlands, May. 12, 2025. DOI: 10.17605/[OSF.IO/QNHSE](https://osf.io/QNHSE).
- Albouy, P., **Kweon, Y.J.**, Whittaker, H., Baillet, S., Zatorre, R.J. (2018). Enhancing learning-related plasticity with information-based neuromodulation. Auditory learning and plasticity symposium. Montreal, Canada, Oct. 23, 2018.